

Reversal of fatigue during prolonged exercise by carbohydrate infusion or ingestion.

Seven cyclists exercised at 70% of maximal O₂ uptake (VO₂max) until fatigue (170 +/- 9 min) on three occasions, 1 wk apart. During these trials, plasma glucose declined from 5.0 +/- 0.1 to 3.1 +/- 0.1 mM (P less than 0.001) and respiratory exchange ratio (R) fell from 0.87 +/- 0.01 to 0.81 +/- 0.01 (P less than 0.001). After resting 20 min the subjects attempted to continue exercise either 1) after ingesting a placebo, 2) after ingesting glucose polymers (3 g/kg), or 3) when glucose was infused intravenously ("euglycemic clamp"). Placebo ingestion did not restore euglycemia or R. Plasma glucose increased (P less than 0.001) initially to approximately 5 mM and R rose (P less than 0.001) to approximately 0.83 with glucose infusion or carbohydrate ingestion. Plasma glucose and R then fell gradually to 3.9 +/- 0.3 mM and 0.81 +/- 0.01, respectively, after carbohydrate ingestion but were maintained at 5.1 +/- 0.1 mM and 0.83 +/- 0.01, respectively, by glucose infusion. Time to fatigue during this second exercise bout was significantly longer during the carbohydrate ingestion (26 +/- 4 min; P less than 0.05) or glucose infusion (43 +/- 5 min; P less than 0.01) trials compared with the placebo trial (10 +/- 1 min). Plasma insulin (approximately 10 microU/ml) and vastus lateralis muscle glycogen (approximately 40 mmol glucosyl U/kg) did not change during glucose infusion, with three-fourths of total carbohydrate oxidation during the second exercise bout accounted for by the euglycemic glucose infusion rate (1.13 +/- 0.08 g/min). (ABSTRACT TRUNCATED AT 250 WORDS)